

Machine Learning Concepts 2

Assignment-2 (Chapter 11 & 12)

Other Types of Learning & Reinforcement Learning

Part A: Theory Questions

1. What is representation learning? Explain its objective and applications.
2. Explain the concept of Independent Component Analysis (ICA). How is it different from PCA?
3. What are autoencoders? Describe their working and objective function.
4. Define active learning. Explain its query strategies with examples.
5. What is instance-based learning? Discuss its characteristics and give examples.
6. Explain ensemble learning and describe any two ensemble techniques such as bagging and boosting.
7. What is Reinforcement Learning? Explain its key characteristics and give real-world examples.
8. What is a policy in Reinforcement Learning? Differentiate between deterministic and stochastic policies.
9. Explain the exploration vs exploitation with suitable examples.
10. What is the Markov Property? Why is it important in Reinforcement Learning?
11. Define a Markov Chain. Explain transition probability with an example.
12. What is a Markov Reward Process (MRP)? Explain its components.
13. What is a value function? Explain the difference between reward and return.
14. Explain the Bellman Equation and its significance in Reinforcement Learning.
15. Differentiate between the following Reinforcement Learning methods:
 - a. Dynamic Programming
 - b. Monte Carlo Methods
 - c. Temporal Difference Learning

Part B: Lab Questions

1. Two source signals are given:

$$S = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 2 \\ 4 & 1 \end{bmatrix}$$

Mixing matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

- a. Generate the mixed signals ($X = AS$).
- b. Apply ICA to recover source signals.
- c. Compare original and recovered signals.

2: Dataset: $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \end{bmatrix}$

- a. Build an autoencoder with architecture $5 \rightarrow 2 \rightarrow 5$
- b. Train the model to reconstruct input
- c. Print encoded (compressed) representation

3: Generate a dataset with • 120 samples • 2 features

- a. Start with 10 labeled samples
- b. Train a classifier
- c. Select most uncertain sample
- d. Add to labeled set
- e. Repeat for 8 iterations

4. Radial Basis Function (RBF)

Given: $X = [1, 3, 5]$; Centers = $[2, 4]$; $\sigma = 1$

- a. Compute Euclidean distance
- b. Apply Gaussian RBF
- c. Calculate hidden layer output

5. Apriori Algorithm:

Transactions:

T1: {A, B}
T2: {A, C}
T3: {B, C}
T4: {A, B, C}

- a. Generate candidate itemsets
- b. Compute support
- c. Find frequent itemsets (min support = 2)

6. ECLAT Algorithm:

Transactions:

T1: {A, B}
T2: {A, C}
T3: {B, C}

- a. Convert to TID list
- b. Compute intersections
- c. Identify frequent itemsets

6. Bagging:

Generate a dataset with 400 samples; 4 features

- Create bootstrap samples
- Train 10 decision tree models
- Perform majority voting
- Compute accuracy

10: Boosting:

Using same dataset as above:

- Train boosting model
- Increase weights for misclassified samples
- Combine weak learners
- Compare performance with Bagging

11: Agent–Environment Interaction

Consider a simple grid environment with states: {0,1,2,3}

- Define possible actions: {Left, Right}
- Simulate agent movement for 10 steps
- Assign rewards (+1 for goal, 0 otherwise)
- Display state transitions

12: Exploration vs Exploitation

- Implement ϵ -greedy strategy with $\epsilon = 0.2$
- Simulate action selection for 20 iterations
- Count number of exploration vs exploitation steps
- Print results

13: Policy Representation

- Define a deterministic policy $\pi(s) \rightarrow a$
- Convert it into a stochastic policy $\pi(s) \rightarrow p(a)$
- Show action probabilities for each state

14: Markov Chain

Weather states: {Sunny, Rainy}

Transition matrix:

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

- Simulate weather for 10 days
- Count frequency of each state

- c. Verify transition probabilities

15: Markov Reward Process

Given:

States: {S1, S2}

Rewards: $R(S1)=5$, $R(S2)=2$

$\gamma = 0.9$

- a. Define transition probabilities
- b. Compute value function for each state
- c. Interpret results

16: Policy Evaluation

Given a simple MDP:

States: {0,1,2}

Actions: {Left, Right}

- a. Initialize value function $V(s)=0$
- b. Perform iterative policy evaluation
- c. Show value updates until convergence

17: Policy Improvement

- a. Use value function from Question 16
- b. Improve policy using greedy approach
- c. Show updated policy

18: Value Iteration

- a. Initialize $V(s)=0$
- b. Apply Bellman optimality update
- c. Run until convergence
- d. Extract optimal policy

19: Monte Carlo Method

- a. Generate 5 episodes of state transitions
- b. Compute returns for each episode
- c. Estimate value function using average returns

20: Q-Learning

- a. Initialize Q-table for states {0,1,2} and actions {Left, Right}
- b. Use learning rate $\alpha = 0.1$ and $\gamma = 0.9$
- c. Apply Q-learning update rule
- d. Train for 20 episodes
- e. Print final Q-table